

melanoma slice · RareShield  
CXCR5-dim TPEX motif surfaced

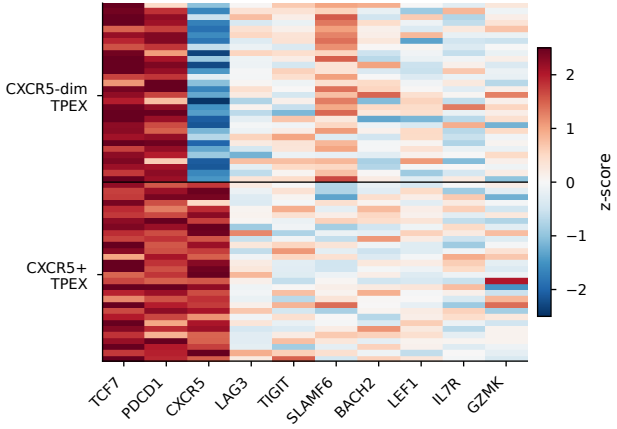
**a**

- CXCR5-dim TPEX
- TCF7+CXCR5+ TPEX

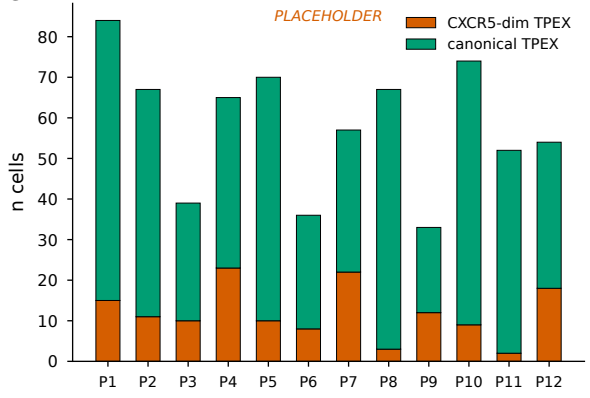
PLACEHOLDER — awaiting Kang-melanoma slice

marker heatmap: motif vs. canonical TPEX

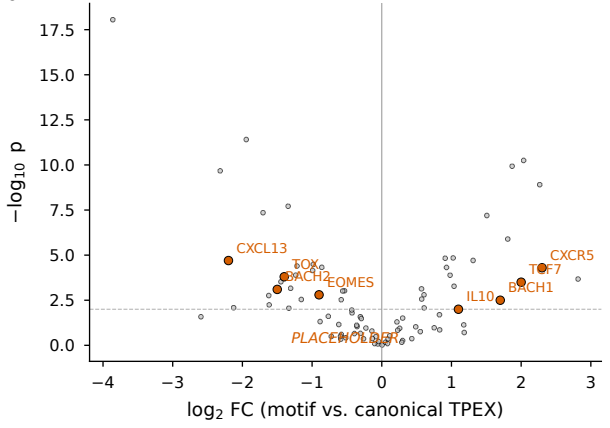
**b**



**c** per-donor consistency (12-donor melanoma cohort)



**d** differential expression vs. nearest canonical subset



**e** therapy-resistance hypothesis

- CXCR5-axis downregulation uncouples from Tfh/germinal-centre niche
  - TCF7+PD1+ retains stem-like program but exits CXCL13 gradient
  - Predicted: diminished response to anti-PD-1 monotherapy
  - Tests: ICI-responder vs non-responder stratification in Bassez 2021 + validation cohort
  - Clinical handle (from Tumor Biology): fragility-weighted score  $\geq 0.6$
- PLACEHOLDER — hypothesis panel; wet-lab hooks after Immunology/Tumor-Bio review